

UFUG2106

Tutorial 01: Review of Chapter 01

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Key Topics

- Variables
- Sets
- Relations and Functions
- Graphs

Quiz 01

Rewrite the statement formally using variables:

“Given any integer, if it is even, then its square is also even.”

- a. Given any integer x , if x is even, then x^2 is also even.
- b. For every integer x , if x is even, then x^2 is also even.
- c. For any integer x , if $x = 2m(m \in \mathbb{Z})$, then $x^2 = 2n(n \in \mathbb{Z})$.
- d. $\forall x \in \mathbb{Z}, x \equiv 0 \pmod{2} \Rightarrow x^2 \equiv 0 \pmod{2}$.
- e. $k \in \mathbb{Z}, x \in 2\mathbb{Z}, x^2 \in 4\mathbb{Z}$
- f. If n is even, then n^2 is also even.

Example 1.1.4 – *Rewriting an Existential Universal Statement*

Fill in the blanks to rewrite the following statement in three different ways:

There is a person in my class who is at least as old as every person in my class.

- a. Some _____ is at least as old as _____.
- b. There is a person p in my class such that p is _____.
- c. There is a person p in my class with the property that for every person q in my class, p is _____.

Example 1.1.4 – *Solution*

- a. person in my class; every person in my class
- b. at least as old as every person in my class
- c. at least as old as q

Ordered n-tuple

The notation for an **ordered n-tuple** generalizes the notation for an ordered pair to a set with any finite number of elements. It also takes both order and multiplicity into account.

Definition

Let n be a positive integer and let $x_1, x_2, \dots, x_n$ be (not necessarily distinct) elements. The **ordered n -tuple**, $(x_1, x_2, \dots, x_n)$, consists of $x_1, x_2, \dots, x_n$ together with the ordering: first x_1 , then x_2 , and so forth up to x_n . An ordered 2-tuple is called an **ordered pair**, and an ordered 3-tuple is called an **ordered triple**.

Two ordered n -tuples $(x_1, x_2, \dots, x_n)$ and $(y_1, y_2, \dots, y_n)$ are **equal** if, and only if, $x_1 = y_1, x_2 = y_2, \dots$, and $x_n = y_n$.

Symbolically:

$$(x_1, x_2, \dots, x_n) = (y_1, y_2, \dots, y_n) \iff x_1 = y_1, x_2 = y_2, \dots, x_n = y_n.$$

Example 1.2.6 – Ordered n -tuples

a. Is $(1, 2, 3, 4) = (1, 2, 4, 3)$?

b. Is $(3, (-2)^2, \frac{1}{2}) = (\sqrt{9}, 4, \frac{3}{6})$?

c. Is $((1,2),3) = (1,(2,3))$?

d. Is $\{\{1,2,3\},\{1,2\},\{1\}\} = (3,2,1)$?

Example 1.2.6 – *Solution*

a. No. By definition of equality of ordered 4-tuples,

$$(1, 2, 3, 4) = (1, 2, 4, 3) \Leftrightarrow 1 = 1, 2 = 2, 3 = 4, \text{ and } 4 = 3$$

But $3 \neq 4$, and so the ordered 4-tuples are not equal.

Example 1.2.6 – *Solution*

continued

b. Yes. By definition of equality of ordered triples,

$$\left(3, (-2)^2, \frac{1}{2}\right) = \left(\sqrt{9}, 4, \frac{3}{6}\right) \Leftrightarrow 3 = \sqrt{9} \text{ and } (-2)^2 = 4 \text{ and } \frac{1}{2} = \frac{3}{6}.$$

Because these equations are all true, the two ordered triples are equal.

Cartesian Products

Definition

Given sets $A_1, A_2, \dots, A_n$, the **Cartesian product** of $A_1, A_2, \dots, A_n$, denoted $A_1 \times A_2 \times \dots \times A_n$, is the set of all ordered n -tuples $(a_1, a_2, \dots, a_n)$ where $a_1 \in A_1$, $a_2 \in A_2, \dots, a_n \in A_n$.

Symbolically:

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n\}.$$

In particular,

$$A_1 \times A_2 = \{(a_1, a_2) \mid a_1 \in A_1 \text{ and } a_2 \in A_2\}$$

is the Cartesian product of A_1 and A_2 .

The Language of Relations and Functions

Definition

Let A and B be sets. A **relation R from A to B** is a subset of $A \times B$. Given an ordered pair (x, y) in $A \times B$, **x is related to y by R** , written $x R y$, if, and only if, (x, y) is in R . The set A is called the **domain** of R and the set B is called its **co-domain**.

The notation for a relation R may be written symbolically as follows:

$x R y$ means that $(x, y) \in R$.

The notation $x \not R y$ means that x is not related to y by R :

$x \not R y$ means that $(x, y) \notin R$.

Exercise

Let $A = \{1, 2, 3\}$, $B = \{u\}$, and $C = \{m, n\}$. Find each of the following sets.

a. $A \times (B \times C)$

b. $(A \times B) \times C$

c. $A \times B \times C$

Example 1.3.2 – *The Circle Relation*

Define a relation C from $\mathbf{R}$ to $\mathbf{R}$ as follows: For any $(x, y) \in \mathbf{R} \times \mathbf{R}$,

$(x, y) \in C$ means that $x^2 + y^2 = 1$.

a. Is $(1, 0) \in C$? Is $(0, 0) \in C$? Is $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \in C$? Is $-2 C 0$? Is $0 C (-1)$? Is $1 C 1$?

b. What are the domain and co-domain of C ?

c. Draw a graph for C by plotting the points of C in the Cartesian plane.

Example 1.3.2 – *Solution*

a. Yes, $(1, 0) \in C$ because $1^2 + 0^2 = 1$.

No, $(0, 0) \notin C$ because $0^2 + 0^2 = 0 \neq 1$.

Yes, $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \in C$ because $\left(-\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{4} + \frac{3}{4} = 1$.

No, $-2 \notin C$ because $(-2)^2 + 0^2 = 4 \neq 1$.

Yes, $0 \in C$ because $0^2 + (-1)^2 = 1$.

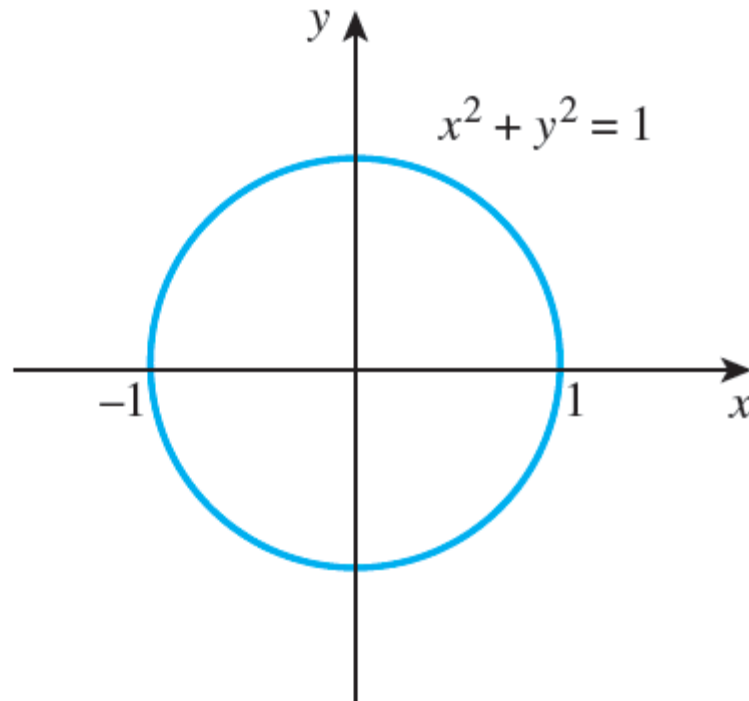
No, $1 \notin C$ because $1^2 + 1^2 = 2 \neq 1$.

Example 1.3.2 – *Solution*

continued

b. The domain and co-domain of C are both $\mathbf{R}$, the set of all real numbers.

c.



Example of Relation:

Reflexive relation: $a R a$.

Symmetric relation: if $a R b$, then $b R a$.

Transitive relation: if $a R b$ & $b R c$, then $a R c$.

Given relation R from S to S , where $S = \{1,2,3\}$

a. How many valid R are there?

Example of Relation:

Reflexive relation: $a R a$.

Symmetric relation: if $a R b$, then $b R a$.

Transitive relation: if $a R b$ && $b R c$, then $a R c$.

Given relation R from S to S , where $S = \{1,2,3\}$. Is the following R reflexive? symmetric? transitive?

- a. $S1 = \{(1,1), (2,2), (3,3)\}$
- b. $S2 = \{(1,2), (2,3), (1,3)\}$
- c. $S3 = \{(1,1), (1,2), (2,1), (2,2), (3,3)\}$
- d. $S4 = \{(2,1), (3,1), (2,3)\}$
- e. $S5 = \{(1,2), (1,3), (2,1), (3,1)\}$
- f. $S6 = ">"$

Example of Relation:

Reflexive relation: $a R a$.

Symmetric relation: if $a R b$, then $b R a$.

Transitive relation: if $a R b$ & $b R c$, then $a R c$.

Equivalence relation: reflexive, symmetric and transitive

Given relation $\sim$ from S to S , where S is _____. Please find some common relation which is _____:

- a. Set of real number R ; Not reflexive, not symmetric, but transitive
- b. Set of real number R ; Reflexive, not symmetric, but transitive
- c. Set of real number R ; Equivalent
- d. Set of the sets; Not reflexive, but transitive
- e. Set of the matrices; Equivalent
- f. Set of the triangles; Equivalent

For any equivalence relation on a set X , the set of its equivalence classes is a partition of X

Example of Relation:

Reflexive relation: $a R a$.

Symmetric relation: if $a R b$, then $b R a$.

Transitive relation: if $a R b$ && $b R c$, then $a R c$.

Equivalence relation: reflexive, symmetric and transitive

Given relation $\sim$ from S to S , where S is _____. Please find some common relation which is _____:

- a. Set of real number R ; Not reflexive, not symmetric, but transitive $<;>$
- b. Set of real number R ; Reflexive, not symmetric, but transitive $\leq; \geq$
- c. Set of real number R ; Equivalent $=; \equiv \text{ mod(on } \mathbb{Z})$
- d. Set of the sets; Not reflexive, but transitive $\subset; \subseteq$
- e. Set of the matrices; Equivalent $\cong B = P^T A P; \sim B = P^{-1} A P; \approx B = P A Q$
- f. Set of the triangles; Equivalent $\cong$ congruent; $\sim$ (similar)

Example 1.3.5 – *Functions and Relations on Sets of Strings*

Let $A = \{a, b\}$ and let S be the set of all strings over A .

- a. Define a relation L from S to $\mathbf{Z}^{nonneg}$ as follows: For every string s in S and for every nonnegative integer n ,

$(s, n) \in L$ means that the length of s is n .

Observe that L is a function because every string in S has one and only one length. Find $L(abaaba)$ and $L(bbb)$.

Example 1.3.5 – *Functions and Relations on Sets of Strings*_{continued}

- b. Define a relation C from S to S as follows: For all strings s and t in S ,

$$(s, t) \in C \text{ means that } t = as,$$

where as is the string obtained by appending a on the left of the characters in s . (C is called **concatenation** by a on the left.)

Example 1.3.5 – *Functions and Relations on Sets of Strings*_{continued}

Observe that C is a function because every string in S consists entirely of a 's and b 's and adding an additional a on the left creates a new string that also consists of a 's and b 's and thus is also in S . Find $C(abaaba)$ and $C(bbb)$.

Example 1.3.5 – *Solution*

a. $L(abaaba) = 6$ and $L(bbb) = 3$

b. $C(abaaba) = aabaaba$ and $C(bbb) = abbb$

The Language of Graphs

The formal definition of a graph follows.

Definition

A **graph** G consists of two finite sets: a nonempty set $V(G)$ of **vertices** and a set $E(G)$ of **edges**, where each edge is associated with a set consisting of either one or two vertices called its **endpoints**. The correspondence from edges to endpoints is called the **edge-endpoint function**.

An edge with just one endpoint is called a **loop**, and two or more distinct edges with the same set of endpoints are said to be **parallel**. An edge is said to **connect** its endpoints; two vertices that are connected by an edge are called **adjacent**; and a vertex that is an endpoint of a loop is said to be **adjacent to itself**.

An edge is said to be **incident on** each of its endpoints, and two edges incident on the same endpoint are called **adjacent**. A vertex on which no edges are incident is called **isolated**.

Example 1.4.7

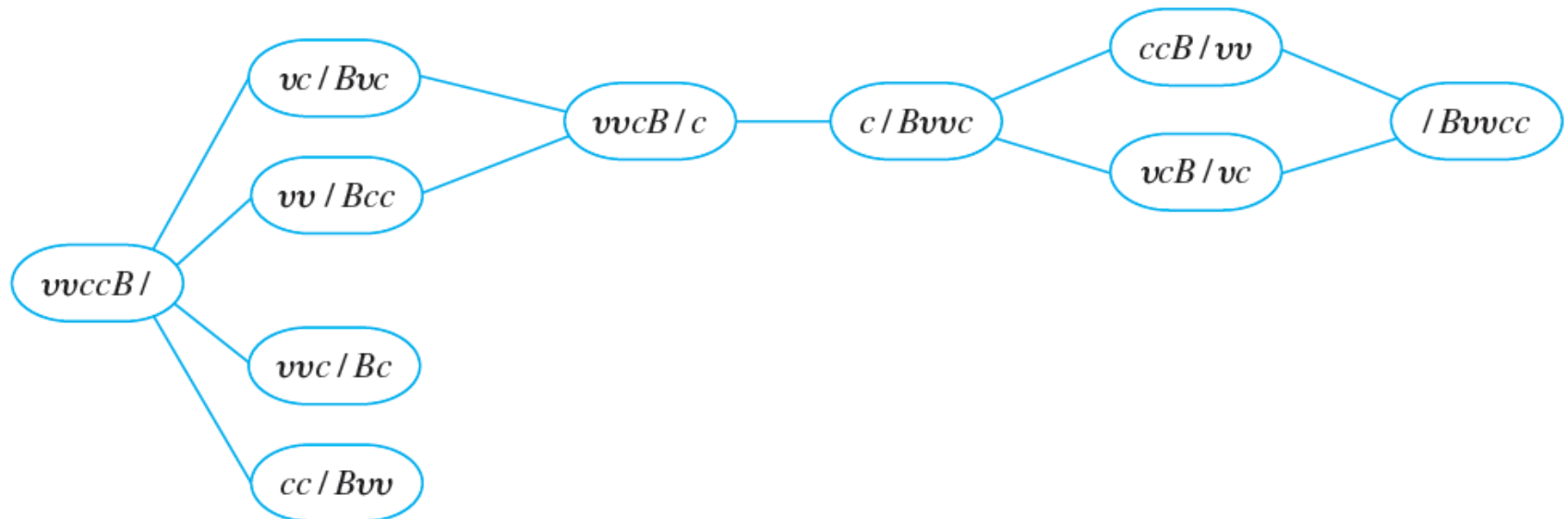
Using a Graph to Solve a Problem: Vegetarians and Cannibals

The following is a variation of a famous puzzle often used as an example in the study of artificial intelligence. It concerns an island on which all the people are of one of two types, either vegetarians or cannibals. Initially, two vegetarians and two cannibals are on the left bank of a river. With them is a boat that can hold a maximum of two people. The aim of the puzzle is to find a way to transport all the vegetarians and cannibals to the right bank of the river. What makes this difficult is that at no time can the number of cannibals on either bank outnumber the number of vegetarians. Otherwise, disaster befalls the vegetarians!

Solution

We could write (vvc / Bc) to indicate that there are two vegetarians and one cannibal on the left bank and one cannibal and the boat on the right bank. Then $(vvccB /)$ would indicate the initial position in which both vegetarians, both cannibals, and the boat are on the left bank of the river.

The aim of the puzzle is to figure out a sequence of moves to reach the position $(/ Bvvcc)$ in which both vegetarians, both cannibals, and the boat are on the right bank of the river.



Exercise 1.4 (12)

Use graph to solve a puzzle: It involves a wolf, a goat, a bag of cabbage, and a ferryman. From an initial position on the left bank of a river, the ferryman is to transport the wolf, the goat, and the cabbage to the right bank. The difficulty is that the ferryman's boat is only big enough for him to transport one object at a time, other than himself. Yet, for obvious reasons, the wolf cannot be left alone with the goat, and the goat cannot be left alone with the cabbage. How should the ferryman proceed?

Solution

To solve this puzzle using a graph, introduce a notation in which, for example, wc / fg means that the wolf and the cabbage are on the left bank of the river and the ferryman and the goat are on the right bank. Then draw those arrangements of wolf, cabbage, goat, and ferryman that can be reached from the initial arrangement ($wgcf /$) and that are not arrangements to be avoided (such as wg / fc)).

At each stage ask yourself, “Where can I go from here?” and draw lines or arrows pointing to those arrangements.

This method gives the graph shown on the right side.

